

ANALYSTLAB AFRICA
Machine Learning Internship Programme

Business Understanding Report
Customer Churn Prediction — Telco Dataset

Week 2 — Data Preprocessing & Feature Engineering for Machine Learning

Programme	Machine Learning Internship
Project	Data Preprocessing & Feature Engineering for Machine Learning
Deliverable	Business Understanding Report
Role	Junior Machine Learning Engineer
Version	1.0

1. Business Objective

AnalystLab Africa's Machine Learning Internship Programme tasks the Junior Machine Learning Engineer with preparing a customer dataset for churn prediction. Telecommunications operators lose recurring revenue every time a customer cancels their subscription — often referred to as "churn". The business goal is not the model itself at this stage, but building a reliable, machine-learning-ready dataset that a future classification model can use to predict which customers are likely to churn.

A model that flags high-risk customers early allows the business to act — offering retention incentives, adjusting contract terms, or improving service quality — before the customer leaves. The value of this project therefore lies in transforming raw, inconsistent operational data into a trustworthy foundation for that prediction.

2. Dataset Overview

The dataset used is the Telco Customer Churn dataset, the same source used in the Week 1 exploratory analysis. It contains one row per customer, covering demographic attributes, subscribed services, contract details, billing information, and the churn outcome.

Property	Value
Rows (raw)	7,043 customers
Columns (raw)	21 features
Target variable	Churn (Yes / No)
Feature types	18 categorical, 3 numerical
Duplicate rows	0

Each row represents a unique customer, identified by customerID, and includes account information (tenure, contract type, payment method), subscribed services (phone, internet, streaming, security add-ons), and billing figures (MonthlyCharges, TotalCharges).

3. Business Questions

The preprocessing pipeline is designed to answer the following questions, as defined in the Week 2 assignment brief:

1. What data quality issues exist in the dataset?
2. Which features require preprocessing?
3. Which encoding techniques are most appropriate?
4. Which scaling method should be applied?

5. Are there significant outliers that need treatment?
6. Which features contribute most to solving the business problem?
7. Is the dataset ready for model training?

Scope note

This report addresses the business framing of these questions. The technical answers — data quality findings, encoding choices, outlier treatment, and feature selection results — are detailed in the companion Data Preprocessing Report.

4. Key Business Drivers Identified

Exploratory work on this dataset points to tenure (how long a customer has been subscribed) as a central driver of churn risk. Customers in their first 12 months show a substantially higher churn rate than longer-tenured customers, which motivated the engineering of a dedicated `tenure_group` feature and an `is_new_customer` risk flag during preprocessing — both intended to make this pattern explicit and easier for a downstream model to exploit.

Contract type and billing structure are also expected to be relevant: month-to-month customers are generally less committed than those on one- or two-year contracts. These relationships are explored quantitatively in the preprocessing phase.

5. Success Criteria for This Phase

- A clean dataset with no duplicate records and no unresolved missing values.
- Correct data types for every column (numeric fields stored as numeric, not text).
- Categorical features encoded and numerical features scaled, ready for a classification algorithm.
- Outliers identified and treated using a documented, reproducible method.
- Every feature-selection decision backed by a quantitative test, not intuition alone.
- A final CSV export that is genuinely model-ready — encoded, scaled, and free of identifier columns.

6. Next Steps

With the business framing established, the Data Preprocessing Report documents the technical execution: data quality findings, cleaning decisions, feature engineering, encoding and scaling choices, outlier treatment, and feature selection — culminating in the machine-learning-ready dataset delivered for this milestone.